CHAPTER 9

Specific Standards for Bond Investment (*Continued*)

THE PROVISIONS OF THE ISSUE

Under this heading come such features as the security of the bonds, the conditions affecting interest payments, and the date of maturity. Conversion and similar privileges, specified in the indenture, are, of course, important in themselves, but they do not enter into the determination of standards for the selection of *fixed-value* investments.

Under the New York statute, only bonds secured by mortgage are eligible in the public-utility group.[1] However, debenture (unsecured) railroad bonds are admitted, provided the earnings and dividend record meet stiffer requirements than are set forth for mortgage issues. The statute also permits the purchase of income bonds (i.e., those on which the obligation to pay interest is dependent upon earnings) on the same basis as debentures.

Obsolete and Illogical Restrictions. In our opinion this set of restrictions is quite out of date and illogical. In view of our emphatic argument in Chap. 6 against attaching predominant weight to specific security, it must be clear that we do not favor the exclusion of any group of unsecured bond issues per se, or even the establishment of any *sharply defined* standards or requirements which favor secured bonds over debentures.

If a company has only one bond issue, it would seem to make little difference whether this is a first mortgage or a debenture, provided the latter is protected against the placing of future issues ahead of it. Needless to say, a debenture bond preceded by a first mortgage is not so attractive as the first-

[1] The specific provisions of the statute are now referred to, without regard to the discretionary powers of the Banking Board to waive any or all of them (*supra* Chap. 8).

mortgage bond itself, even though the investor's chief reliance in both cases must be the same—*i.e.*, the ability of the company to meet all its obligations. But this distinction would be equally applicable to a secondmortgage issue and hence is not concerned with debentures as such. We have already discussed the practicalities of selecting as between senior and junior liens (Chap. 6) and shall refer to this point again when we consider interest coverage.

Income Bonds in Weaker Position than Debentures. While the New York statute is too severe in its categorical exclusion of all unsecured public-utility issues, its acceptance of railroad *income* issues on the same basis as railroad *debentures* is fully as objectionable for the opposite reason. The provisions of income bonds vary greatly among the different issues, the basic distinction being between those on which interest must be paid if earned and those over which the directors have a greater or lesser measure of discretion. Generally speaking, income bonds are allied more closely to preferred stocks than to ordinary fixed obligations. We shall consider them, accordingly, in our chapter on preferred stocks, in which we shall set forth the need for especial caution and strictness in the selection of this type of security for *straight investment*.

Standards of Safety Should Not Be Relaxed Because of Early Maturity. Investors are inclined to attach considerable importance to the *maturity date* of an issue, because of its bearing on whether it is a short- or long-term security. A short maturity, carrying with it the right to repayment soon after purchase, is considered an advantageous feature from the standpoint of safety. Consequently, investors are prone to be less exacting in their standards when purchasing notes or bonds due in a short time (say, up to three years) than in their other bond selections.

In our opinion this distinction is unsound. A near maturity means a problem of refinancing for the company as well as a privilege of repayment for the investor. The bondholder cannot count on the mere fact of maturity to assure this repayment. The company must either have the cash available (which happens relatively seldom) or else an earning power and financial position which will permit it to raise new funds. Corporations frequently sell short-term issues because their credit is too poor at the time to permit of a long-term flotation at a reasonable rate. Such a practice frequently results in trouble for the company, and therefore for the investor, at maturity.

Examples: The Fisk Rubber Company sold $10,000,000 of five-year 5½s in 1926. In 1929 they sold at 96 because of their near maturity, although the company's earnings exhibit was unsatisfactory. But payment of principal was defaulted at maturity in 1931; the company went into receivership; and the price of the notes fell to 10¾ in that year.

In 1929 the New York, Chicago and St. Louis Railway (Nickel Plate) sold $20,000,000 of three-year 6% notes. They have been repeatedly extended but only with great difficulty and upon threat of insolvency if the holders refused to extend. (In 1936 they sold as low as 26¾.)

A recent example of apparently unwarranted partiality accorded by the bond market to an issue of near maturity is supplied by Pennsylvania-Dixie Cement Company First 6s, due September 1941, which in early 1939 sold above par. This issue had barely covered interest charges (on a reduced depreciation basis) in 1937 and 1938, and had reported deficits in the six preceding years. Net current assets were less than the outstanding bonds. It was possible, of course, that conditions in 1941 might permit the repayment of this security; but those who bought it at a full price in 1939 were undoubtedly taking an unnecessary risk of severe shrinkage of principal value.

Distinctions between Short and Long Maturities of the Same Issue. There have been quite a number of cases in which investors have been willing to pay much higher prices for a short-term issue than for an equally secured long-term issue of the same company. In nearly every case this has proved a mistake—because either (1) the company's credit improved, in which case the distant maturity had a much greater rise in price, or else (2) the company was unable to pay off the short term issue at maturity.

Examples of (1):

	Low Price 1932
Lehigh Valley Coal Company First Refunding 5s, due 1934	96½
Lehigh Valley Coal Company First Refunding 5s, due 1944	35

The company was able to pay off the 1934 issue at maturity, but in the meantime the 5s of 1944 had advanced to 91. See also example under (2) below.

	Low Price 1932
U. S. Rubber Company Secured 6½s, due 1933	94
U. S. Rubber Company Secured 6½s, due 1935	43

The 1933 issue was paid at maturity, but so was the 1935 issue, which of course proved by far the better purchase.

Examples of (2):

	High Price 1938
Lehigh Valley Coal Company First Refunding 5s, due 1944	99⅞
Lehigh Valley Coal Company First Refunding 5s, due 1954	45

Interest was defaulted in January 1939, and the price of the 1944 issue collapsed to 36, versus 20 for the 1954 maturity.

	Low Price 1932
Pressed Steel Car Debenture 5s, due 1933	82
Pressed Steel Car Debenture 5s, due 1943	40

Both defaulted on Jan. 1, 1933, and were ultimately treated alike in the reorganization.

	High Price 1934
Standard Gas and Electric Debenture 6s, due 1935	82
Standard Gas and Electric Debenture 6s, due 1951	40

The company failed to meet the 1935 maturity. In the ensuing reorganization the various debenture issues were treated practically alike, and in 1939 they all sold at the same price.

Because of the foregoing discussion and examples we advise against the drawing of distinctions between long- and short-term issues such as

result in any relaxation of standards of safety in the selection of issues of the latter type.[2]

RECORD OF INTEREST AND DIVIDEND PAYMENTS

Bonds purchased on an investment basis should have behind them a sufficiently long record of successful operation and of financial stability on the part of the issuer. New enterprises and those recently emerged from financial difficulties are not entitled to the high credit rating essential to justify a fixed-value investment.[3] A similar disqualification would logically apply to states or municipalities which have failed to meet their obligations punctually at any time over a preceding period of years.

Provisions of New York Statute. The New York statute recognizes this criterion and gives it concrete expression as follows: Bonds of states other than New York are eligible if the state has not defaulted on interest or principal payments during the previous *ten years*. For municipalities outside New York State, the period is *twenty-five years*; for railroads, *six years*; for gas, electric, and telephone companies, *eight years*.

With respect to bonds of corporations, however, the requirements as to earnings coverage to be discussed under the next heading—should adequately take care of the question of past record. The time covered by the earnings requirement is only a little shorter than the periods above suggested, and hence it would seem an unnecessary complication to exact a past-solvency test in addition to an earnings test.

Civil obligations, on the other hand, are not sold on the basis of an earnings record. Consequently the investor is compelled to attach primary importance to a satisfactory history of punctual payment. The

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[2] In an exceptional case a short-term issue may be bought at an investment price, even though the earnings exhibit is inadequate, provided the working capital position is so strong as to assure payment without difficulty. Such an investment would correspond to a loan made by a commercial bank.

Example: This would apply to Central Steel Company First 8s, assumed by Republic Steel Corporation, due Nov. 1, 1941, and selling in November 1939 at 109 to yield 3.31%. Note also that preference may properly be given to short maturities at times as a matter of *investment policy*, but not to the extent of relaxing the standards of safety.

[3] This statement might not apply in those cases in which the financial difficulties were due to an excessive debt burden which the reorganization reduces to a figure that would have been amply taken care of by the previous earnings.

requirement on this point set forth in the New York statute would no doubt appear reasonable to the average investor.

We cannot recommend such a rule of investment, however, without considering the results that would follow from its general adoption. If all purchases of municipal bonds required a clean record for 25 years, how could any township float a bond issue during the first quarter-century of its existence? And similarly, if a state or city has been driven into default, how will it finance itself during the 10 or 25 years, respectively, needed to restore its obligations to the eligible list? In the case of corporations, such financing might be accomplished on a speculative basis, through the sale of stock, or convertible bonds, or even bonds at a large discount. But such methods are not open to municipalities. The difficulty is met in actual practice by raising the coupon rate on the obligations of states or municipalities with inferior credit. For example, a city emerging from financial embarrassment might be able to attract new funds by offering a 5% coupon rate in contrast with 2% paid by New York State.[4] But this solution of the problem runs counter to the principle, previously developed, that a high coupon rate is not adequate compensation for the assumption of substantial risk of principal. In other words, it would be a mistake to buy a municipal obligation for its high yield, if it is recognized as inferior in grade and subject to more than a nominal possibility of default.

A Dilemma and a Suggested Solution. We are faced therefore by a dilemma, since the theoretically correct attitude of the bond buyer would render impossible the necessary financing of many municipalities. Viewing the matter realistically, it may be dismissed with the observation that there will always be enough undiscriminating investors on hand to absorb the bonds of any town or village which offers a seemingly attractive rate. Consequently the logical and careful bond buyer can avoid such issues without fatal results to borrowers having second-rate credit.

This disposition of the dilemma is too cynical to be entirely satisfactory. The ideal solution would probably lie in setting up some especially stringent quantitative tests to compensate for the failure by a municipality to meet the twenty-five year requirement of punctual payment. If a city

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[4] Note that in November 1939 City of Detroit obligations due 1954 (which had been in default in 1933) sold at a 3.70% yield basis, as against a return of about 2% on similar bonds of smaller municipalities with a good record.

has fallen into financial difficulties, it must rehabilitate itself by reducing its expenditures, or by raising its tax rate and other revenue, or possibly by a compulsory scaling down of its debt, corresponding to a corporate reorganization. By such means the town may place its finances on an entirely new and sound basis entitling it to a satisfactory credit rating in spite of its previous default. But the prudent investor will accord such a credit rating only after a careful study of the financial exhibit, including such items as the relation of expenditures and total debt, on the one hand, to population, property values and revenues, on the other. The bond buyer should expect to obtain a higher than standard yield on municipal obligations of this character, in repayment *not for the assumption of special risk, but for the effort required to satisfy himself of the soundness of the issue.*

A similar attitude should be taken towards newly organized civil bodies, where only a short record of debt service is available.[5]

The Dividend Record. The statutes governing legal investments have traditionally laid great stress upon a satisfactory record of dividend payments by the issuing enterprise. In most states a bond is eligible only if the company has paid regular dividends in certain minimum amounts for at least five years. This requirement is evidently based on the theory that since corporations exist in order to pay dividends, only those which do in fact pay dividends may be said to be really successful and therefore suitable for bond investment.

Dividend Record Not Conclusive Evidence of Financial Strength. It may not be denied that dividend-paying concerns as a class are more prosperous than non-dividend payers. But this fact would not in itself justify the summary condemnation of all the bonds of non-dividend-paying enterprises. An exceedingly strong argument *against* such a rule lies in

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[5] The technique of analysis of state or municipal finances is elaborate and it does not lend itself to dependable short cuts. An adequate treatment of the subject would lie outside the purview of this book or the competence of the authors. We refer the reader to treatments of the subject in standard works on investment such as Hastings Lyon, Investment, pp. 56–179, New York, 1926; Ralph E. Badger and Harry G. Guthmann, *Investment: Principles and Practices*, pp. 735–780, rev. ed., New York, 1936; and to *Proceedings of the Conference on Bond Portfolios* 1939 of the New York State Bankers Association, pp. 136–158, New York, 1939; *Investment Standards and Procedure*, which is *Commercial Bank Management Booklet* No. 19, issued by the Bank Management Commission of the American Bankers Association, New York, 1937; A. M. Hillhouse, *Municipal Bonds: A Century of Experience*, New York, 1936.

the fact that the payment of dividends is only an indication of financial strength; and not only does it fail to afford any direct advantage to the bondholder, but it may often be injurious to his interests by reducing the corporation's resources. In actual practice the dividend provisions of the statutes governing legal investments have at times had consequences directly opposite to those intended. Railroad companies in a weak financial position have improvidently continued dividend payments for the particular purpose of maintaining their bonds on the eligible list, so that the very practice supposed to indicate strength behind the bond has in reality undermined its safety.[6]

The Role of the Dividend Record in Bond Investment. The evidence given by the balance sheet and income account must be regarded as a more dependable clue to the soundness of an enterprise than is the record of dividend payments. It seems best therefore to dispense with all hard and fast rules on the latter point in determining the suitability of bond issues for straight investment. But the failure of a company to pay dividends when the earnings appear satisfactory should properly cause an intending bond buyer to scrutinize the situation with more than usual care, in order to discover whether the policy of the directors is due to weak elements in the picture not yet reflected in the income account. We might also point out incidentally that the bonds of dividend-paying companies possess a certain mechanical advantage in that their owners may receive a definite and perhaps timely warning of impending trouble by the later passing of the dividend; and being thus placed on their guard, they may be able to protect themselves against serious loss. Bonds of non-dividend-paying concerns are at a certain disadvantage in this respect, but in our opinion this may be adequately offset by the exercise of somewhat greater caution on the part of the investor.

The New York statute is somewhat more progressive than those of other states in its treatment of the dividend question. Railroads are

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[6] Cf. the testimony of the chairman of the New Haven in December 1936, in the Interstate Commerce Commission's investigation of that road, admitting that dividends were paid in 1931 to keep its bonds "legal" and listing other roads that paid unearned dividends presumably for the same reason (see *New York Times* of Dec. 3, 1936). For a much earlier example, see Dewing's discussion of the payment of unearned dividends by Boston and Maine Railroad in 1911–1913, to keep its bonds legal (*Financial Policy of Corporations*, 3d rev. ed., p. 609*n*). Also see our reference to the Wabash-Ann Arbor in 1930, note 10 in Chap. 25.

required alternatively either to have paid dividends of a certain amount in five out of the last six years, or failing this, to meet more stringent requirements as to coverage of fixed charges. Public-utility companies are required either to have paid certain dividends in each of the five preceding years, or else to have earned an amount equal thereto. This provision falls into the error of the other statutes by possibly impelling payment of unearned dividends. The progressive idea appears in the converse side of the provision, which waives payment of dividends so long as they are earned.

RELATION OF EARNINGS TO INTEREST REQUIREMENTS

The present-day investor is accustomed to regard the ratio of earnings to interest charges as the most important specific test of safety. It is to be expected therefore that any detailed legislation governing the selection of bond investments would be sure to include minimum requirements in respect to this cardinal factor. Nevertheless the majority of the statutes cover this point in only a fragmentary and inadequate manner. The legislatures have relied to a considerable extent on their requirements as to the company's dividend record to assure a satisfactory earning power.[7] As we have just pointed out, this criterion is open to serious objection. The superiority of the New York statute is manifest chiefly in two provisions: first, its recognition of the prime importance of an adequate earnings record; and secondly, its consistent treatment of a company's *total* fixed charges as an indivisible unit.

Requirements of the New York Law. The requirements of the New York law with respect to *earnings coverage* may be summarized as follows:

In the case of railroad-mortgage bonds (or collateral-trust bonds equivalent thereto) and railroad-equipment obligations, the company must have earned its fixed charges 1½ times in five out of the six years immediately preceding, and also in the latest year. If dividends have not been paid as stipulated, then the period is set at nine out of the ten preceding years.

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[7] Vermont, for example, permits investment in bonds of New England railroads without any earnings test; in the case of other roads the fixed charges must not exceed 20% of the *gross* business. A record of continuous dividend payments is required in both cases.

In the case of other kinds of railroad bonds, e.g., debentures, income obligations, etc., the fixed charges (plus interest on income bonds, if any) must be earned twice in both the latest year and in five out of the six preceding years. In this category, the requirement as to dividend payments is apparently absolute, and no substitute therefor is admitted.

In the case of gas, electric, and telephone bonds, the *average* earnings for the past five years must have equalled twice the *average* total-interest charges, and the same coverage must have been shown in the latest year.

Three Phases of the Earnings Coverage: 1. Method of Computation. In analyzing these statutory provisions, three elements deserve consideration. The first is the *method* of computing the earnings coverage; the second is the *amount* of coverage required; and the third is the period required for the test.

The Prior-Deductions Method. Various methods are in common use for computing and stating the relation of earnings to interest charges. One of these (which may be called the Prior-deductions Method) is thoroughly objectionable. Nevertheless, prior to 1933 it was followed by the majority of issuing houses in their circulars offering junior bonds for sale, because it makes for a deceptively strong exhibit. The procedure consists of first deducting the prior charges from the earnings and then calculating the number of times the junior requirements are covered by the balance. The following illustration will show both the method itself and its inherent absurdity:

Company A has $10,000,000 of first-mortgage, 5% bonds and $5,000,000 of debenture 6% bonds.

Its average earnings are	$1,400,000	
Deduct interest on first 5s	500,000	earned 2.8 times
Balance for debenture 6s	$ 900,000	
Interest on debenture 6s	$ 300,000	earned 3 times

A circular offering the 6% debenture issue was likely to state that "as shown above" the interest charges are covered three times. It should be noted, however, that the interest on the first 5s is covered only 2.8 times. The implication of these figures would be that the junior issue is better protected than the senior issue, which is clearly absurd. The fact is that the results shown for junior bonds by this prior-deductions method are completely valueless and misleading. One of the favorable results of the

Securities Act of 1933 has been the abandonment of this indefensible method of stating interest coverage in new bond offerings. This change has been due, apparently, not to any specific prohibition by the statute or the S.E.C. regulations but rather to the desire to avoid risking penalties for deceit.

Some Canadian bond offering circulars still use the prior-deductions method. *Example:* Famous Players Canadian Corporation, Ltd., First and Collateral Trust Bonds, Series A, offered about June 1936.

The Cumulative-Deductions Method. The second procedure may be called the Cumulative-Deductions Method. Under this method, interest on a junior bond is always considered in conjunction with prior and equivalent charges. In the example given, the interest on the debenture 6s would be computed as earned $1^3/_4$ times, found by dividing the combined charges of *both* issues, namely $800,000, into the available earnings of $1,400,000. The first-mortgage interest, however, would be said to be earned 2.8 times, since bond interest *junior* to the issue analyzed is left out of consideration in this method. The majority of investors would regard this point of view as entirely sound, and the procedure has been specifically prescribed by a number of states in their enactments governing the eligibility of bonds for savings-bank investment.[8]

The Total-Deductions or "Over-All" Method. In a previous chapter, however, we have emphasized the primary importance of a company's ability to meet all its fixed obligations, because insolvency resulting from default on a junior lien invariably reacts to the disadvantage of the prior-mortgage bondholders. An investor can be sure of his position only if the total-interest charges are well covered. Consequently, the conservative and therefore advisable way of calculating interest coverage should always be by the "totaldeductions method"; *i.e.*, the controlling figure should be the number of times that *all* fixed charges are covered. This would mean that the *same earnings ratio would be used in analyzing all*

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[8] See, for example, Maine, Sec. 27, Chap. 57 of Revised Statutes, as amended by Chap. 222 of Public Laws 1931, subsections VI, VII and VIII, dealing with obligations of steam railroads, public utilities and telephone companies. Similar provisions are to be found in the Vermont statute relative to public-utility bonds. New Hampshire permits the cumulativedeductions method for railroad and public-service company bonds; but, rather strangely, it requires the total-deductions method in the case of the bonds of telephone and telegraph companies.

the fixed interest bonds of any company, whether they are senior or junior liens. In the example above given, the ratio would be $1^3/_4$, as applied to either the first 5s or the debenture 6s. In bond circulars and annual reports this method is now commonly referred to as the "over-all basis" for computing interest coverage.[9]

It is important to bear in mind that fixed charges exclude income-bond interest which is a *contingent* charge. The words "interest charges" and "bonded debt" are also used, for convenience, to refer only to fixed interest bonds unless the context indicates otherwise.

There is no reason, of course, why the coverage for a senior bond should not be computed by the cumulative-deductions method also, and if this coverage is very large it may properly be regarded as an added argument in favor of the issue. But our recommendation is that in applying any *minimum* requirement designed to test the company's strength, the total fixed charges should always be taken into account. The New York statute holds consistently to this very stand, and in our opinion it deserves to be approved and followed.

2. Minimum Requirements for Earnings Coverage. The preference accorded by the New York statute to railroad bonds over public-utility issues is no longer justified, and the more recent record of both groups suggests that their relative positions should be reversed. It is necessary, also, to add a minimum figure for industrial bonds, which should clearly be set higher than for either utilities or rails. Taking these factors into account, we should recommend the following minimum requirements for the coverage of total fixed charges:

Public utilities	$1^3/_4$ times
Railroads	2 times
Industrials	3 times

[9] The phrases: "earnings ratio," "times interest earned," and "earnings coverage," all have the same significance. The statement that "interest is covered $1^3/_4$ times" is more readily understood than the equivalent expression, sometimes used, that "the factor of safety is 75%," and we should advise the consistent use of the former type of expression. Some authorities (e.g., Moody's "*Manual of Investments*" prior to 1930) have used the expression "margin of safety" to mean the ratio of the balance after interest to the earnings available for interest. Example: If interest is covered $1^3/_4$ times the margin of safety becomes $^3/_4 \div 1^3/_4 = 42^6/_7\%$.

3. The Period Comprised by the Earnings Test. Our summary of the New York provisions regarding earnings coverage pointed out that the five-year average is used in the case of utility issues. For railroad bonds, however, the stipulated minimum margin must be shown in five separate years out of the latest six. In all instances, the minimum must be met in the year immediately preceding the date of investment.

Requirements such as the last two are easy to promulgate, but they are poorly suited to the realities of bond investment in an economic world subject to recurring years of serious depression. If it should be characteristic of business in general to experience eight prosperous or average years followed by two unprofitable ones, the effect of these rules would be to encourage investment in bonds (at high prices) during good times, and to impel their sale (at low prices) during depressions.[10]

In our view, the only practical *rigid* application of a minimum-earnings standard must be to the *average* results over a period of time. A five-year average, as prescribed by the statute in the case of public-utility bonds, would seem too short under many circumstances, and we should suggest a *seven-year period* as a more suitable normal standard. But this might be shortened somewhat to exclude clearly abnormal years. (For example, the six-year period 1934–1939 would probably provide a fairer test period than the seven-year period 1933–1939.)

If the test had been made, say, in 1934 or 1935, it would have been better to use a ten- or even twelve-year period to avoid giving undue weight to years of severe depression. Practical considerations suggest also that averaging-in the large deficits experienced by some industrial companies during 1931–1933 might produce an earnings-coverage figure too low to be fairly representative of the current situation, even though a long-term average were taken. This difficulty may be solved, arbitrarily, by considering the earnings in deficit years as zero instead of the actual negative figure.

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[10] The impracticability of these provisions of the New York statute is best evidenced by the fact that annual amendments were deemed necessary between 1931 and 1937 inclusive, their effect being to exclude the results of 1931 through 1936 from the earnings test. This moratorium" terminated in April 1938, at which time over $3,000,000,000 par value of railroad bonds were removed from the eligible list. A new moratorium retains bonds of carriers that have earned interest charges *once over* in the last year and in five out of the last six years.

Example: Interest coverage of Fairbanks Morse Company Debenture 4s, due 1956, as of early 1938.

Interest charges, 1937	$232,000
Earned after interest and taxes, 1937	2,148,000
1937 interest earned	10.2 times
Total earnings after interest, 1928–30 and 1934–1937	11,740,000
Total deficits after interest, 1931–1933	8,873,000
Annual earnings after interest, 1928–1937	287,000
Indicated 10-year coverage for 1937 interest charges	2.2 times
Alternative basis for calculating the 10-year coverage:	
10-year average earnings after interest, counting 1931–1933 years as zero	$ 1,174,000
Revised 10-year coverage for 1937 interest charges	6.1 times
Stock-equity ratio	$3.42 of stock at market for each $1 of bonds at par

The second, or revised, average must be considered as a more realistic reflection of the company's earning power than the straight ten-year average, which fails to meet our minimum requirement. We trust, however, that from 1940 on it will be possible to use seven-year averages, or longer, without having to meet a similar problem.

Other Phases of the Earnings Record. There are, of course, a number of other aspects of the earnings picture to which the investor would do well to pay attention. Among these are the *trend*, the *minimum* figure, and the *current* figure. The importance of each of these cannot be gainsaid, but they do not lend themselves effectively to the application of hard and fast rules. In this case, as in the matter of mortgage security previously discussed, a distinction must be drawn between the few factors which can successfully be embraced by definite and universally applicable rules, and the many other factors which resist such exact formulation but must nevertheless be taken into account by the *judgment* of the investor.

Unfavorable Factors May Be Offset. The practical method of dealing with elements of the latter type may be illustrated in this matter of the earning exhibit. The investor must *demand* an average at least equal to the minimum standard. In addition, he will be *attracted* by: (a) a rising trend of profits; (b) an especially good current showing; and (c) a satis-

factory margin over interest charges in *every* year during the period studied. If a bond is deficient in any one of these three aspects, the result should not necessarily be to condemn the issue but rather to exact an average earnings coverage well in excess of the minimum and to require closer attention to the general or qualitative elements in the situation. If the trend has been unfavorable, or the latest figure alone has been decidedly poor, the investor should certainly not accept the bond unless the average earnings have been substantially above the minimum requirement—*and unless also he has reasonable grounds for believing that the downward trend or the current slump is not likely to continue indefinitely*. Needless to say, the amount by which the average must be advanced in order to offset an unfavorable trend or current exhibit is a matter within the discretion of the investor to determine, and cannot be developed into any set of mathematical formulas.

The Relation of the Coupon Rate to the Earnings Coverage. The theory of earnings coverage is complicated by the arithmetical fact that this coverage varies inversely with the rate of interest. Given the same earnings, interest on a 3% bond issue would be earned twice as many times as it would be if the rate were 6%. Consider the following comparison:

	Utility Company *A*	**Utility Company *B***
Earnings for interest	$600,000	$600,000
Interest charges	(3% on $10,000,000)	(5½% on $10,000,000)
	300,000	550,000
Times interest earned	2.00	1.09

The difference in coupon rates alone makes Company *A* pass our earnings coverage test, whereas Company *B* barely earns its interest. This point may well raise several questions, *viz.*: (1) Can a bond be considered "safe" merely because it carries a low coupon rate? (2) What would be the effect on this safety of a rise in the general rate of interest? (3) Are the bonds of Company *A* a sounder purchase for investment than those of Company *B*? Let us attempt to answer these questions briefly in their order.

1. Effect of Coupon Rate on Safety. Safety, in the technical sense of assurance of continued payment of interest, can certainly be created or destroyed by varying the coupon rate. It is not feasible to think of a 5½%

bond as being safe as to 3% interest and unsafe as to the additional 2½%. Safety of interest is an indivisible concept and must apply to the entire interest charge, the reason being that inability to pay part of the contractual interest—or even junior interest—will result in financial difficulties. These in turn mean the destruction, at least temporarily, of the investment status.

Safety in the sense of maintenance of principal value can also be "created" by a low rate of interest, provided this rate is considered to be permanent—*i.e.*, lasting either through maturity or for a great many years in the future. If the 3% rate is permanent, the earnings of $600,000 should enable Company *A* to refund its bonds at maturity, and they should also maintain the market price of the bonds not far from par.

Allowance must be made for the fact that the rate of interest tends to vary inversely with the ability of the company to pay it. A strong company borrows at a low rate, although it could afford to pay more than could a weak company. This means that "good credit" itself produces "better credit" through its own saving in interest charges, whereas the opposite is equally true. Although this may seem paradoxical and unfair, it must be accepted as a fact in security analysis.

2. Effect of a Rise in Interest Rates on Safety. A general rise in interest rates would not affect the ability of a company to meet its interest charges during the life of its low-rate bond issue. But if they mature in a short time, it will be faced with the problem of refunding at a higher rate, to effect which its earnings must show an adequate margin above this higher rate. On the other hand, if the maturity is distant the market price of this and other bonds will decline substantially should the general rate of interest experience a considerable rise. (Note that the Dow-Jones Index of bond prices declined about 30% between 1917 and 1920, reflecting a rise in interest rates.)

It follows, therefore, that safety of principal, in the sense of maintenance of market value, is certain to be affected adversely in the case of long-term bonds by a sharp rise in the rate of interest.[11] Safety of principal of short-term debt *may* be affected adversely by such a rise in interest rates if the earnings coverage does not exceed our minimum by a comfortable margin.

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[11] An exception would be high-coupon bonds whose price had been held down by a callable feature.

The practical conclusion must be that if the investor considers a rise in interest rates probable, he should not buy long-term low-coupon bonds, no matter how strong the company; and he should buy short-term issues only if earnings would cover a higher coupon rate with an adequate margin. If, however, he is convinced that the low interest rates are here to stay, he may accept them in the same way as the higher rates were formerly accepted. If he is undecided as to the future of interest rates, the best policy might seem to be to confine purchases to bonds of fairly short maturity (say not longer than ten years) and also to increase his earnings coverage requirement to offset the low coupon rate.

3. Relative Attractiveness of the Two Bonds. Our third question relates to the comparative attractiveness of the 3% bonds of Company A and the 5½% bonds of Company *B*. In strict logic the 5½% bond must certainly be more desirable than the 3% bond, since the 5½% bondholder could always place his claim to the extra 2½% on a contingent basis and thus make his company's margin above fixed charges the same as Company *A*'s. But in practice such a reduction of fixed interest is likely to be made only after the issuer has fallen into financial difficulties, which in turn would cause a substantial decline in the market price of the issue. Hence, as a practical matter, it is possible that the holder of the 3% bond may fare better than the owner of the 5½% bond.

However, the anomaly evident in our example should carry a warning to the investor not to pay about par for a 3% bond on the showing of Company *A* unless he is absolutely convinced of the permanence of very low interest rates. (It will also indicate that there are certain speculative opportunities inherent in a bond of the Company *B* type if it is selling at a very low price because of the small margin above its high interest charges—especially if continuance of low interest rates is expected.)